

ENAMEL

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Lecture by Prof Farzeen/Enamel/2022

HYDROXY APPETITE CRYSTALS

- The principal mineral component of enamel is calcium hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$ in crystalline form.
- Hydroxyapatite comprises about 88–90% of the tissue by volume, corresponding to about 95–96% by weight
- The mineral content increases from the enamel–dentine junction to the surface.
- Tooth shade is affected by crystal size and carbonation.

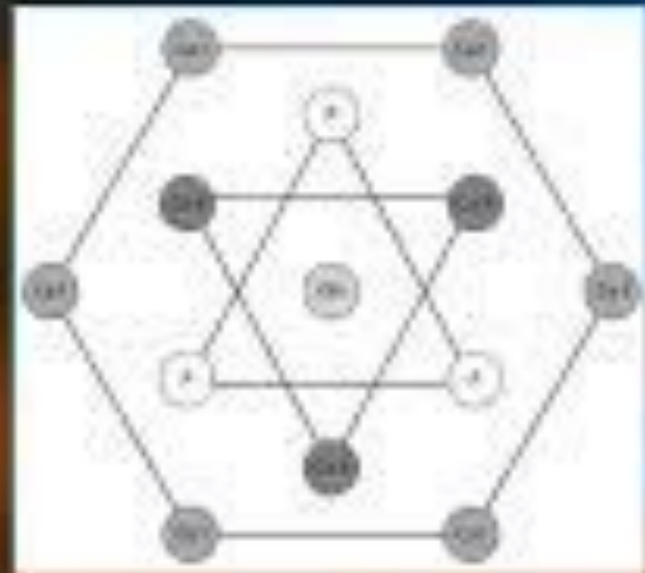
Hydroxyapatite Crystals

- Enamel hydroxyapatite crystals are about 70nm in width, 25nm thick and of great length (almost the full thickness of enamel).
- Most crystallites are hexagonal in cross section.
- The cores of the crystals are richer in magnesium and carbonate in comparison to the peripheries.

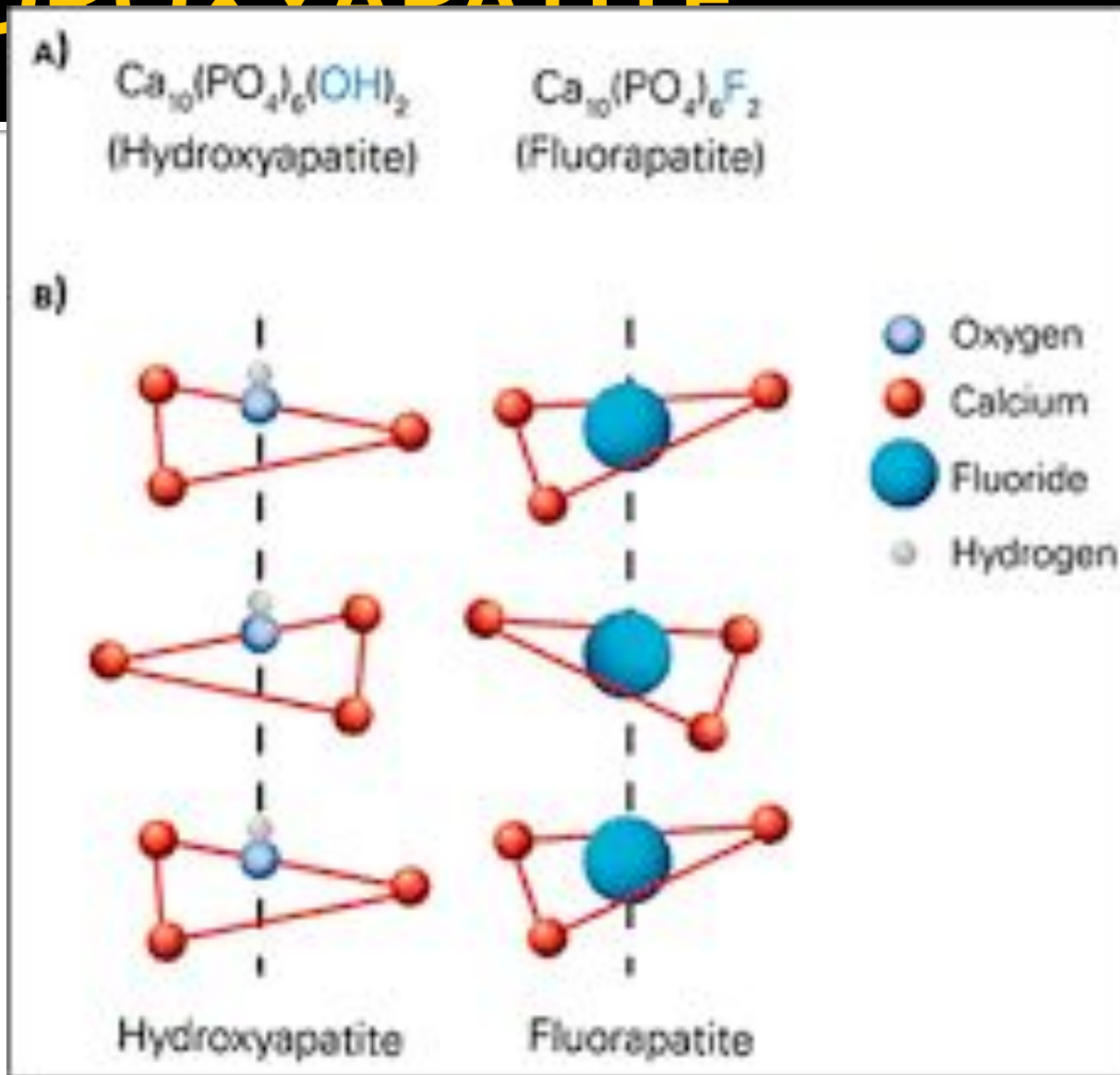


Hydroxyapatite Crystals

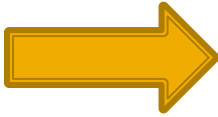
- Each crystal unit has a hydroxyl group surrounded by 3 calcium ions which are surrounded by 3 phosphate ions. Six calcium ions in a hexagon enclose the phosphate ions.
- The crystal is made of a repetition of those planes of ions side by side in stacked layers.



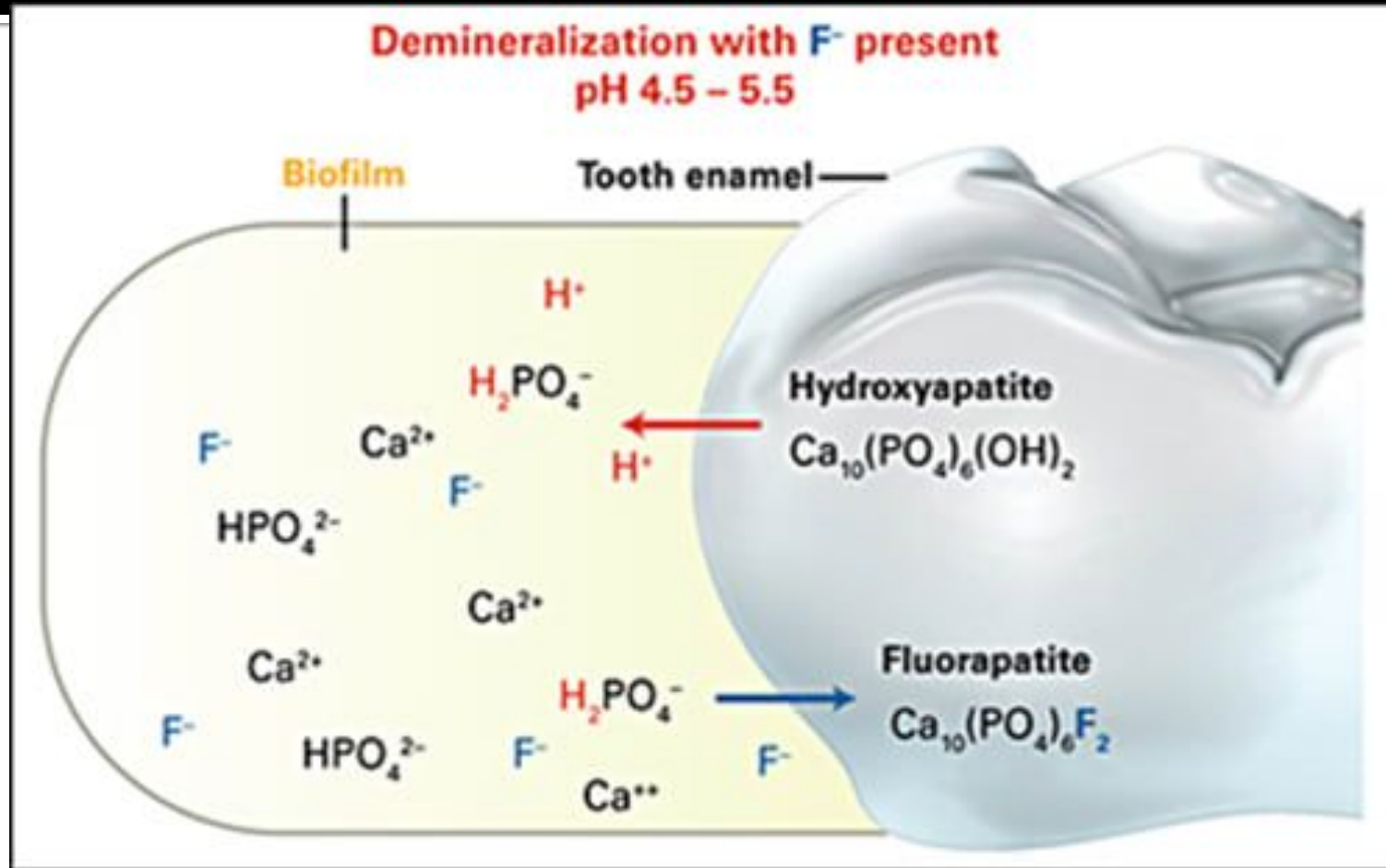
HYDROXYAPATITE



HYDROXYAPATITE

- The basic molecular arrangement  highly organised, it is subject to variation.
- 'Normal' ions may be replaced by different ionic species.
- Carbonate  phosphate / hydroxyl site (about 90% being found at the phosphate position).
- Magnesium  calcium ion

MINERAL DIFFUSION!!!!



FLOURIDE

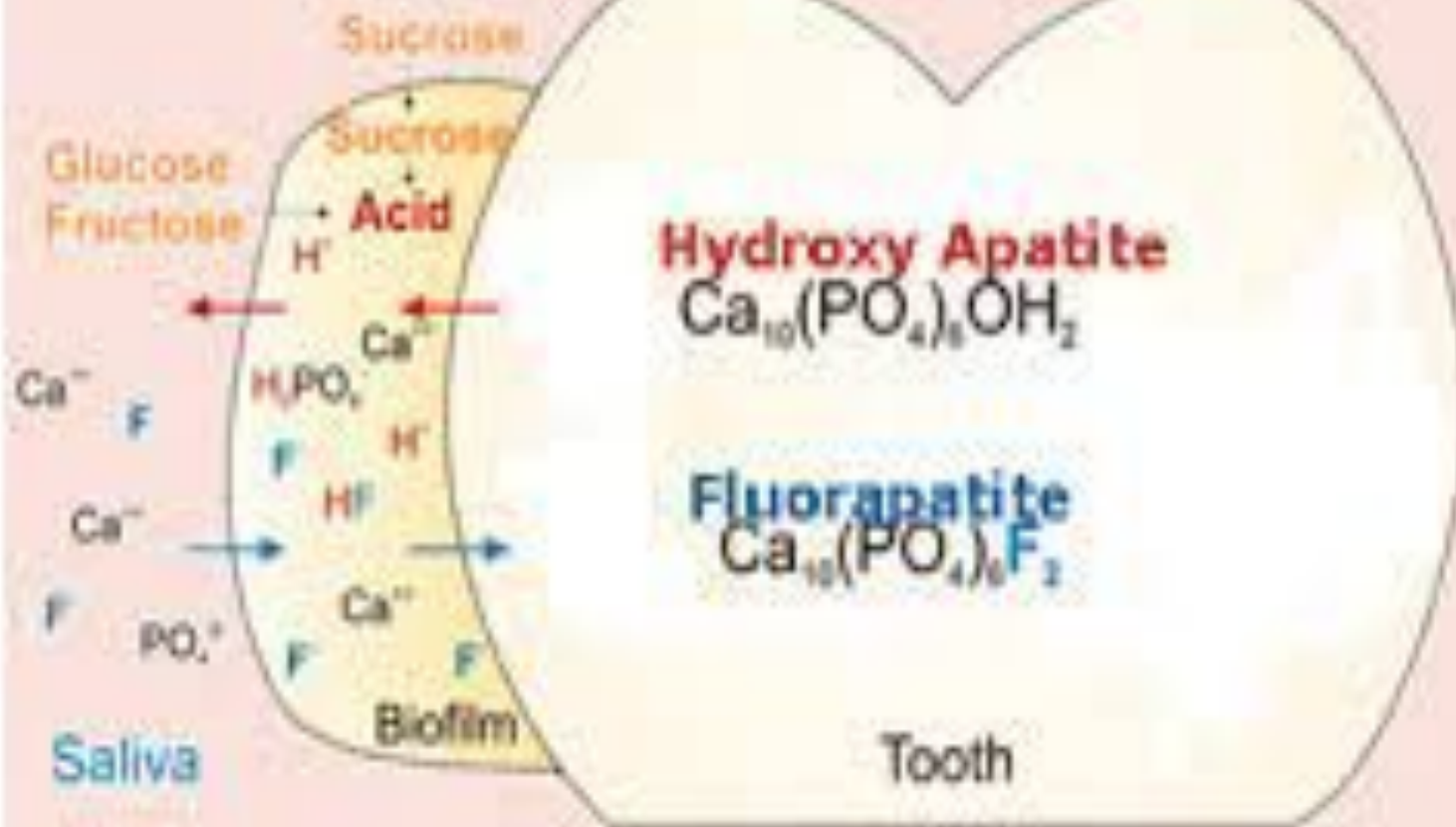
- Fluoride  hydroxyl ions, ↑ stability
- and resistance to acidic dissolution.
- Fluoride levels (unlike magnesium
- and carbonate) **decline from the outer surface towards the dentine**, perhaps because the fluoride is acquired during enamel maturation.

FLUORIDE IN ENAMEL HOApetite

- Exchange of ions with saliva at the surface is important clinically, particularly relating to fluoride.
- Fluoride also becomes more concentrated as the enamel–dentine junction is reached.
- Enamel crystals increase in width from the enamel–dentine junction to the surface zone.
- The crystals are tightly packed with gaps (pores) of <5 nm between them which would be occupied by organic matrix and/or water

Demineralization + F

pH < 5.5; pH > 4.5



WATER

- Water constitutes about 2% by weight of enamel, corresponding to 5–10% by volume. The presence of water is related to the porosity of the tissue.
- Some of the water may lie between crystals and surround the organic material, some may be trapped within defects of the crystalline structure and the remainder forms a hydration layer coating the crystals. The water component and its distribution are of clinical importance because ions such as fluoride travel through it

ORGANIC MATRIX



- mechanical properties of enamel, such as fracture toughness.
- The organic component of range from 0.05% to 3%.(depending on organisation of crystal).
- The organic matrix is made up of peptide groups termed amelogenins and non-amelogenins.
- In the adult matrix, less than 10% is composed of amelogenins, whereas, in developing enamel, 90% is composed of amelogenins.
- amelogenins are selectively reabsorbed during enamel maturation

Lipid material

cross-striations, enamel striae (of Retzius), Hunter–Schreger bands ,in the prism sheaths & prism cores of mature human

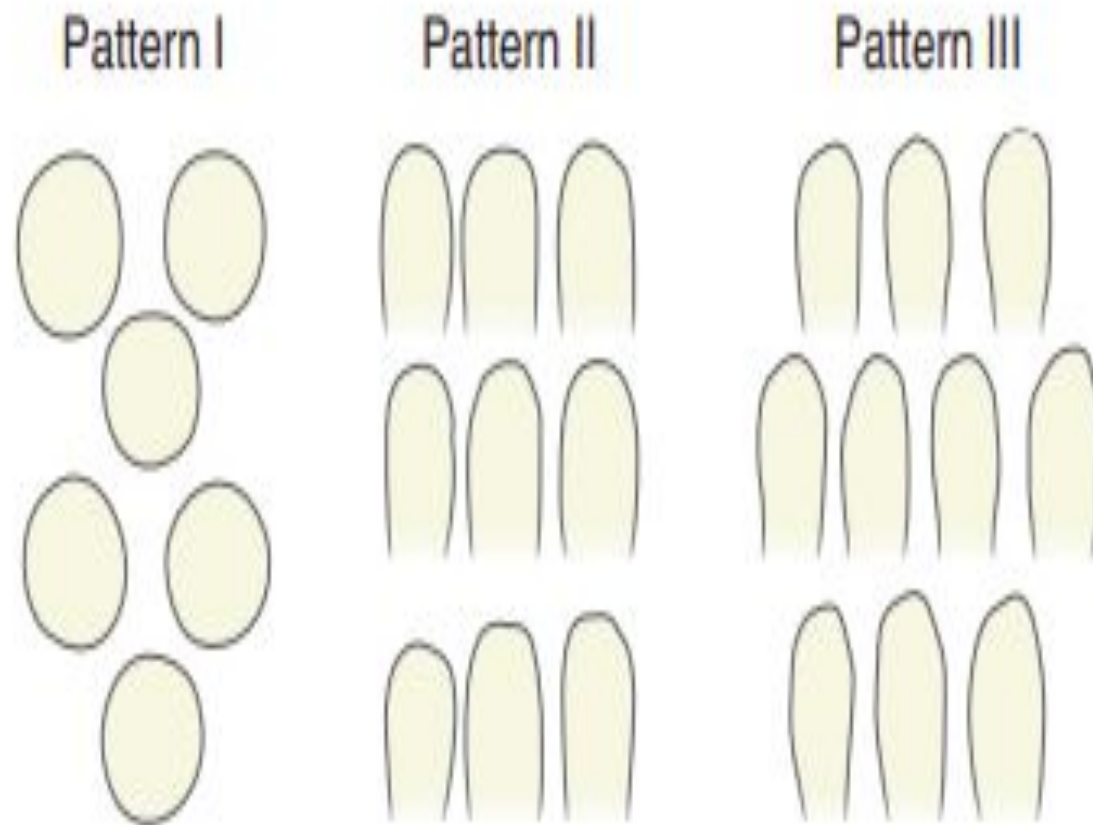


Fig. 7.10 The three prism patterns seen in human enamel. In pattern I enamel the prisms are circular. In pattern II enamel the prisms are aligned in parallel rows. In pattern III enamel the prisms are arranged in staggered rows such that the tail of a prism lies between two heads in the next row, giving a keyhole appearance.

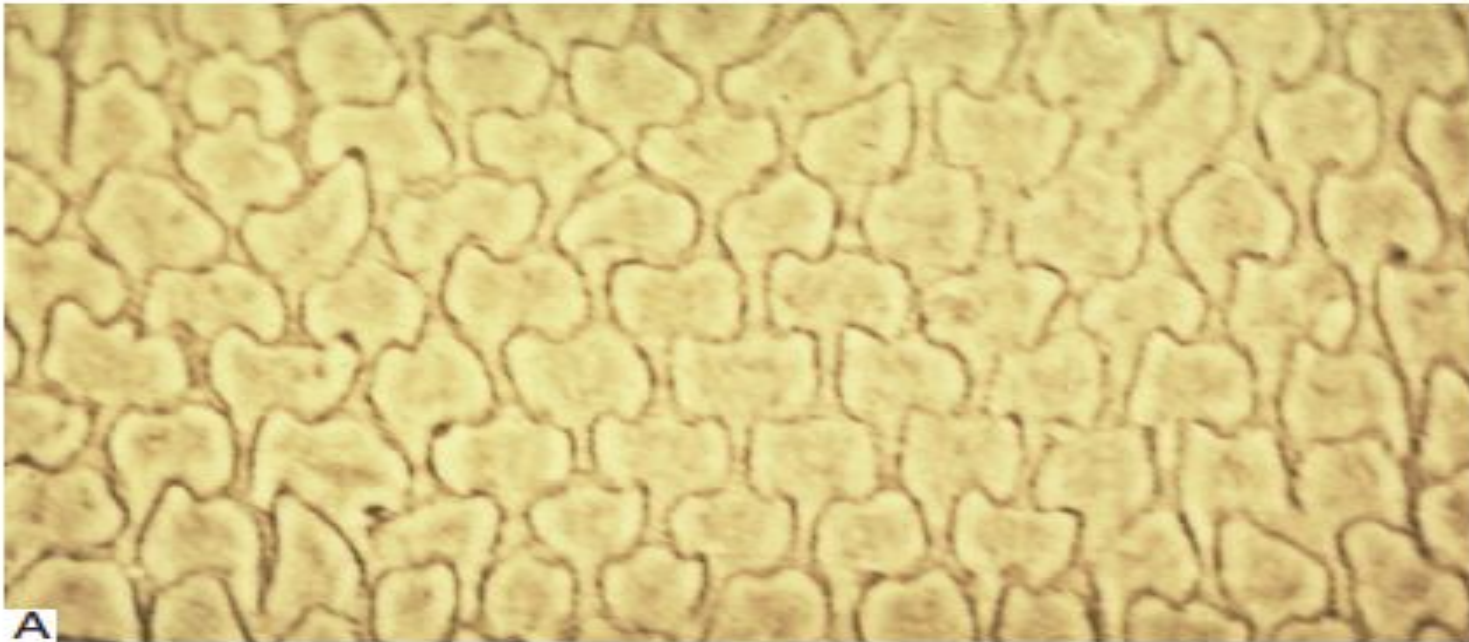
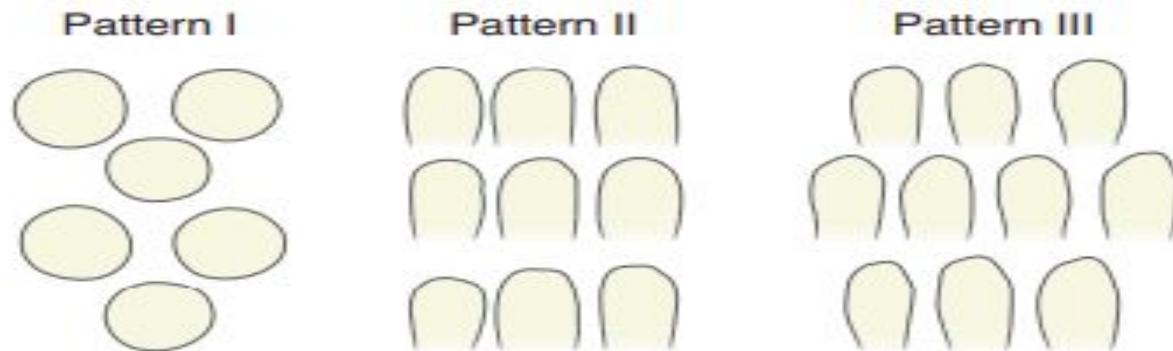


Fig. 7.9 (A) Enamel prisms in transverse section demonstrating the keyhole pattern seen in most regions surface. The lines running obliquely (A) are enamel striae ($\times 250$). Courtesy Dr D F G Poole.



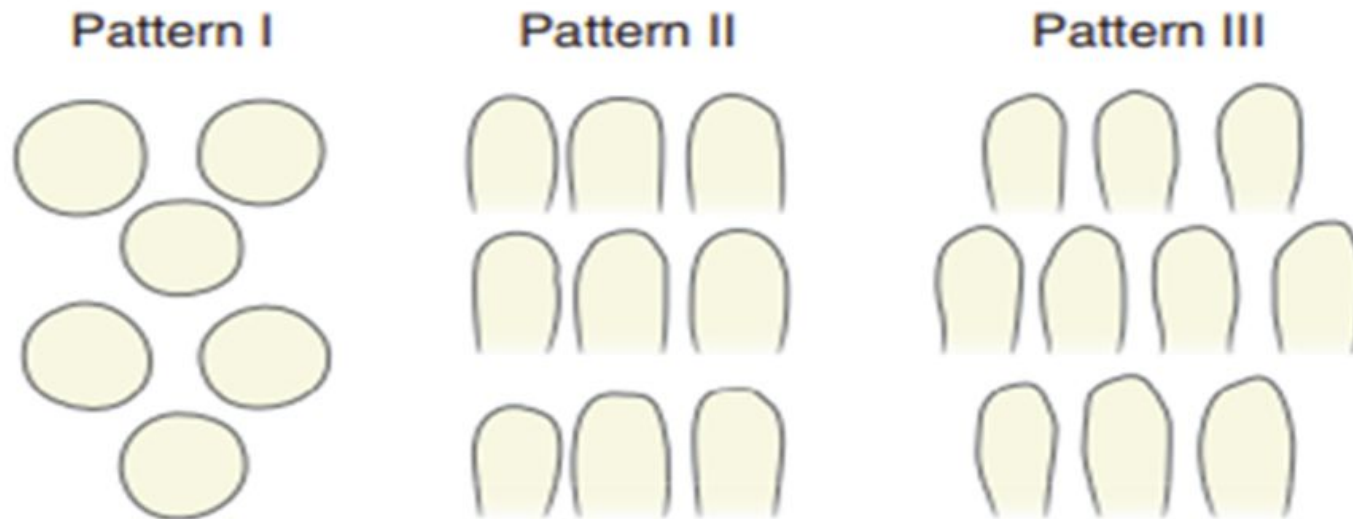
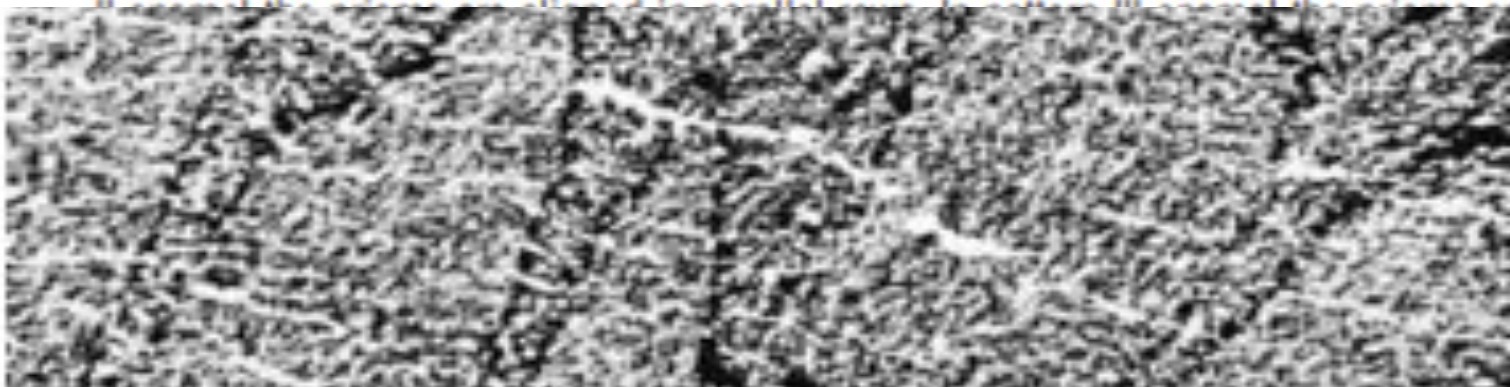
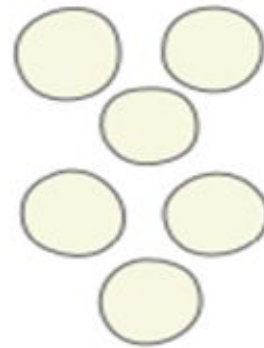


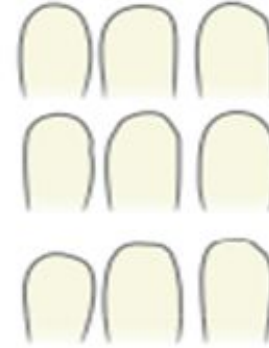
Fig. 7.10 The three prism patterns seen in human enamel. In pattern I enamel the prisms are arranged in a non-regular, scattered pattern. In pattern II enamel the prisms are arranged in a regular, parallel pattern. In pattern III enamel the prisms are arranged in a more complex, irregular pattern.



Pattern I



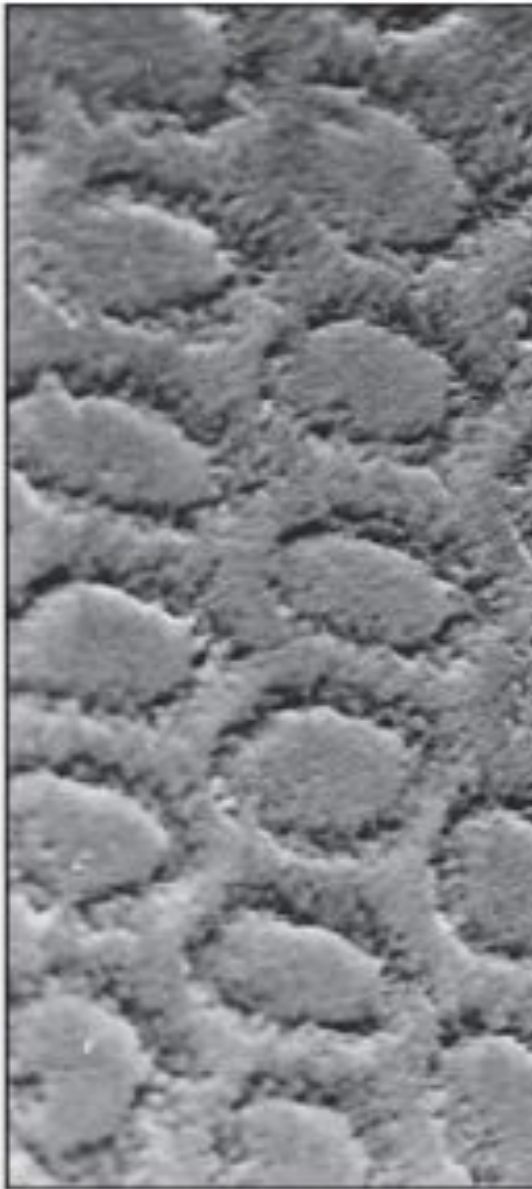
Pattern II



Pattern III



Fig. 7.10 The three prism patterns seen in human enamel. In pattern I enamel prisms are arranged in a hexagonal pattern, in pattern II enamel prisms are arranged in a rectangular pattern, and in pattern III enamel prisms are arranged in a rectangular pattern.



ENAMEL PRISM PATTERN

- In cross-section, the shape of an enamel prism approximates to one of three main patterns `
- Pattern I prisms have complete boundaries and there is a clear distinction between prisms and interprismatic enamel.
- In patterns II and III the prism
- boundary is incomplete and in the cervical region there is a gradual change in orientation between the intraprismatic and interprismatic crystals.

- Between 10 and 13 layers of prisms follow the same direction but blocks above and below follow paths in different directions .
- These periodic changes in prism direction give rise to a banding pattern termed the Hunter–Schreger bands (Fig. 7.23).
- These bands are approximately 50 μm wide and are visible because the different bands of prisms reflect or transmit light in different directions

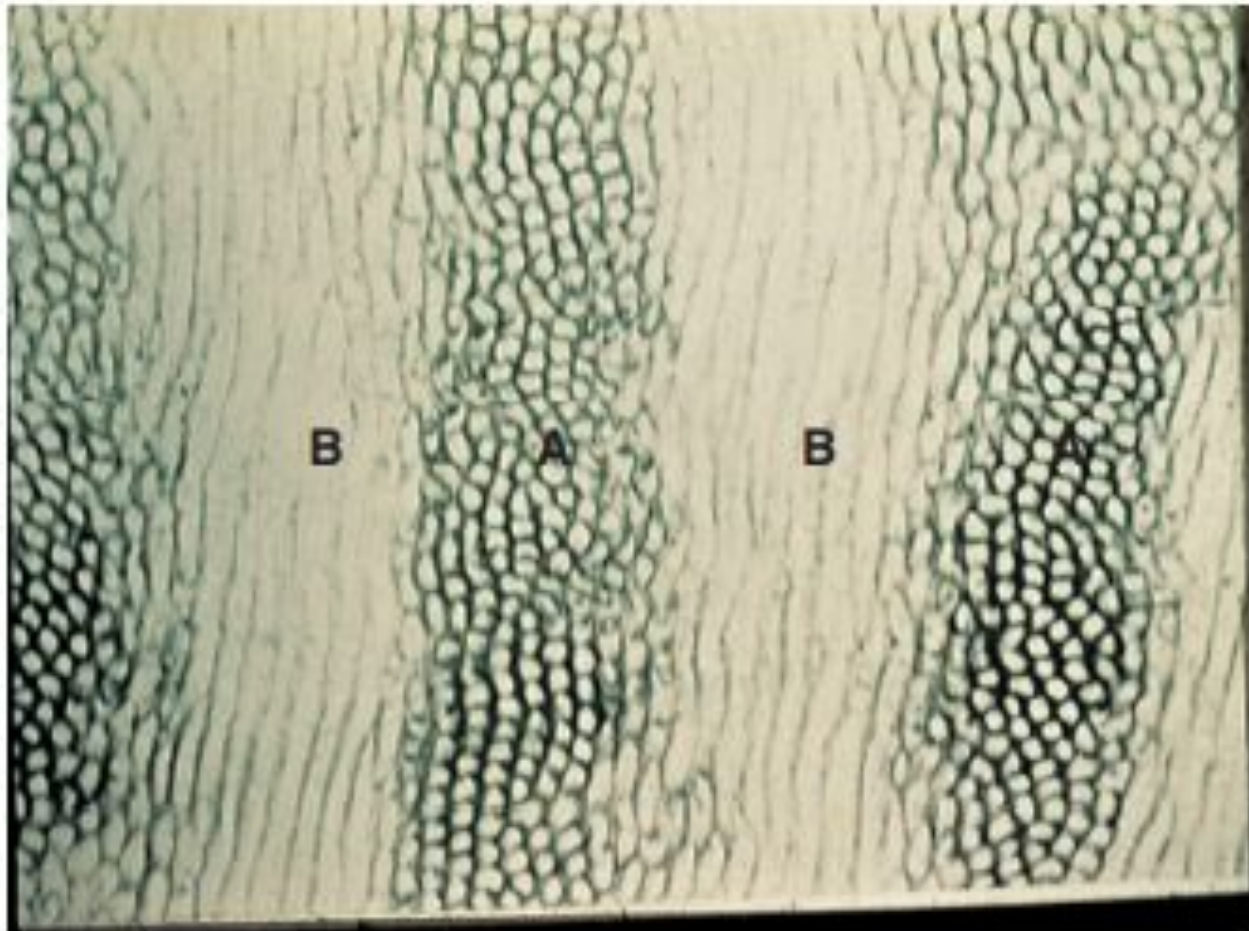
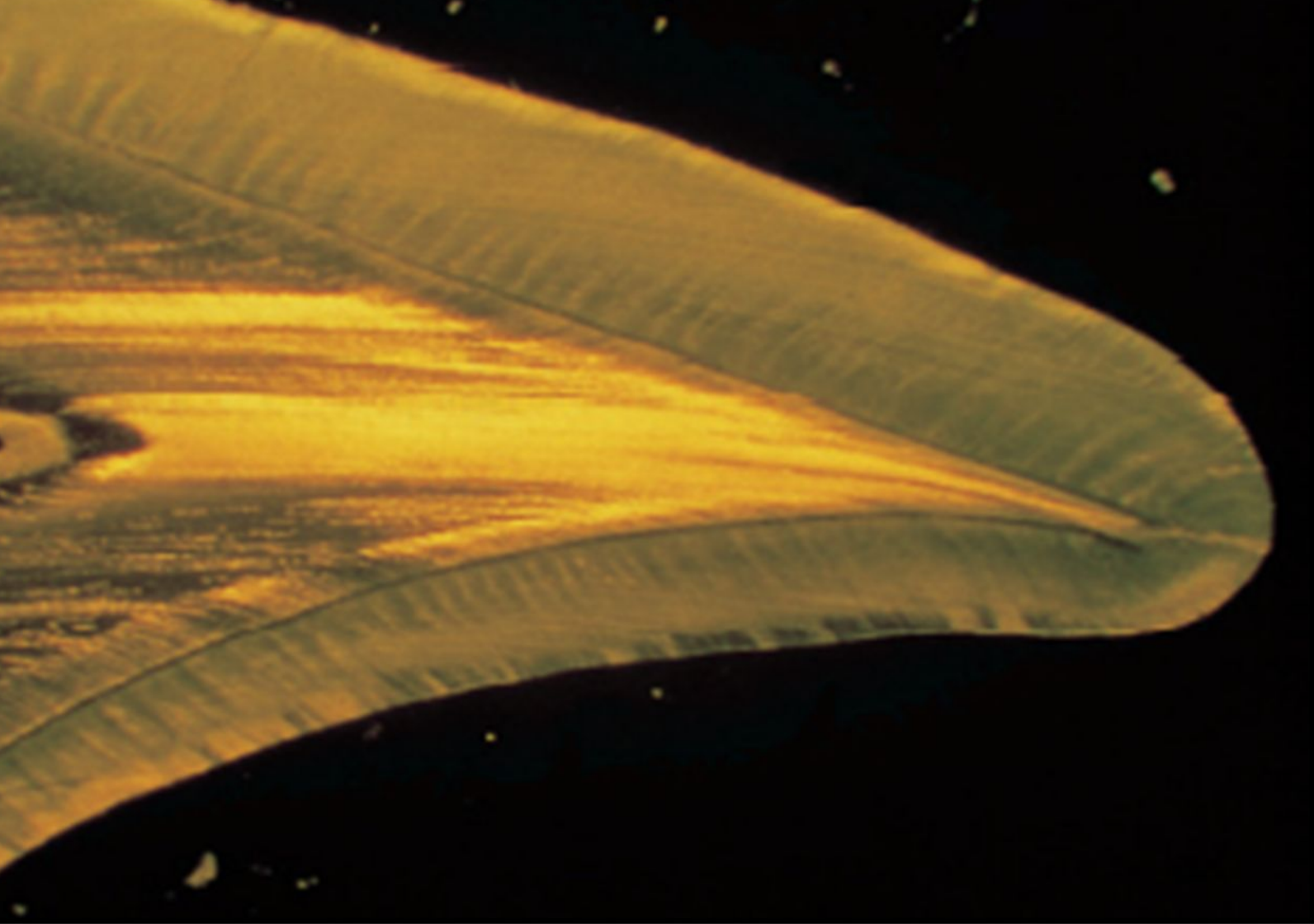
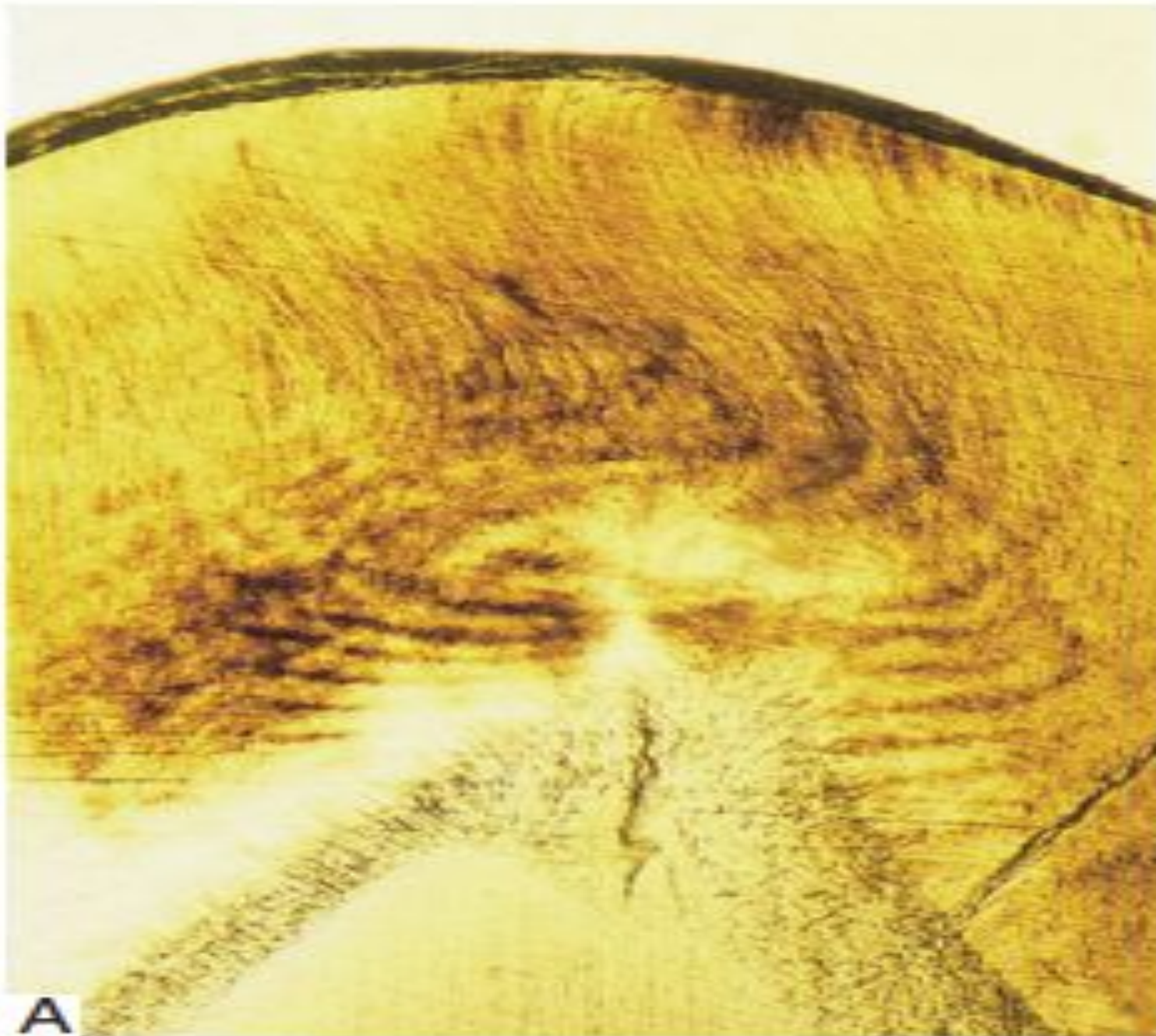


Fig. 7.22 Longitudinal section of enamel showing alternating regions with groups of prism sectioned more transversely (A) or more longitudinally (B) that give rise to the appearance of Hunter–Schreger bands ($\times 30$). Courtesy Dr B A W Brown.

- The bands of prisms that are cut longitudinally are known as parazonies and those cut transversely as diazonies.
- The angle between parazonies and diazonies is about 40 degrees.
- This complex pattern of prisms makes enamel resistant to fracture and, when exposed on the surface, leads to a microridged grinding surface

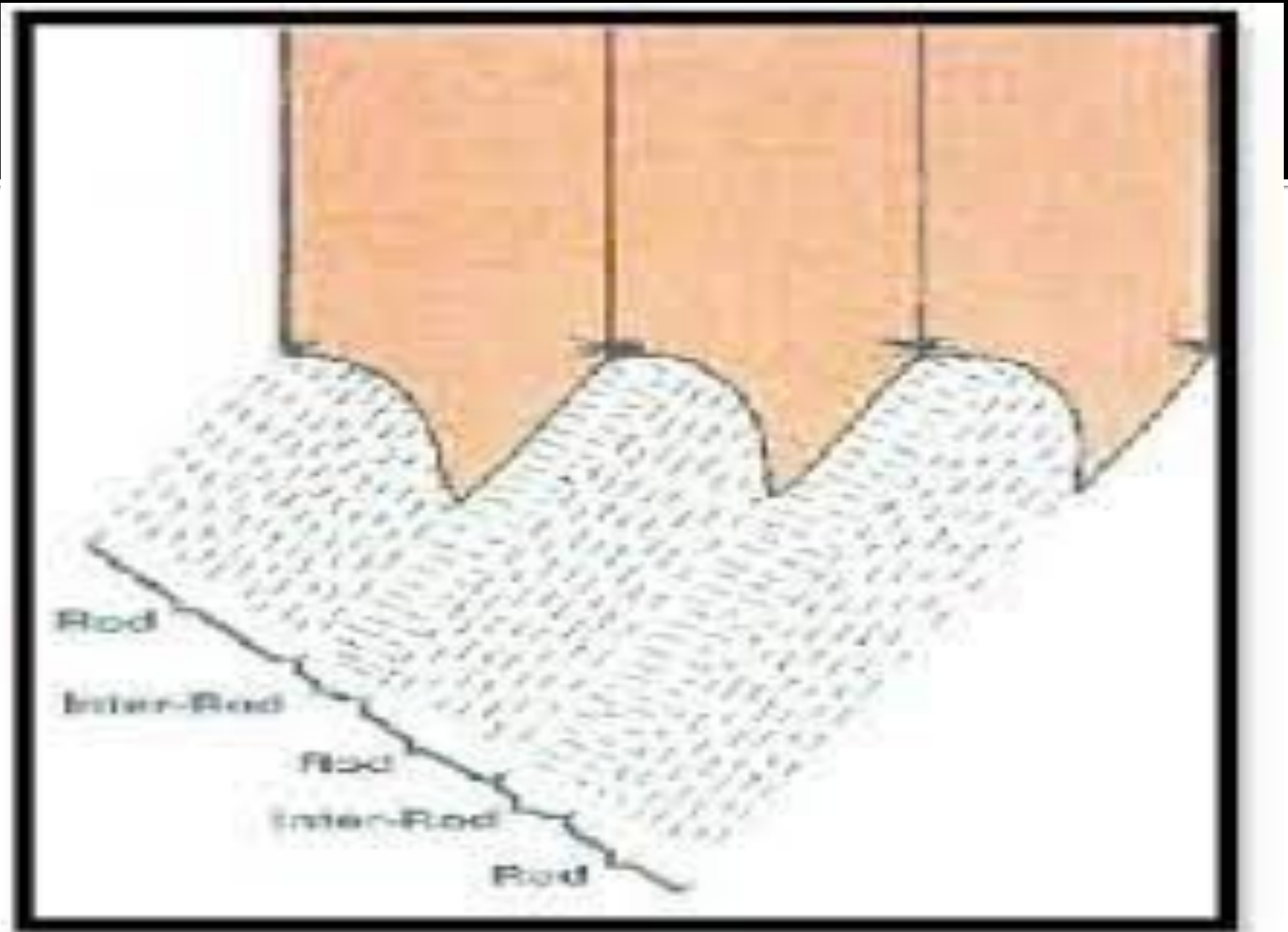


Longitudinal section of enamel showing Hunter–Schreger bands in reflected light ($\times 100$)

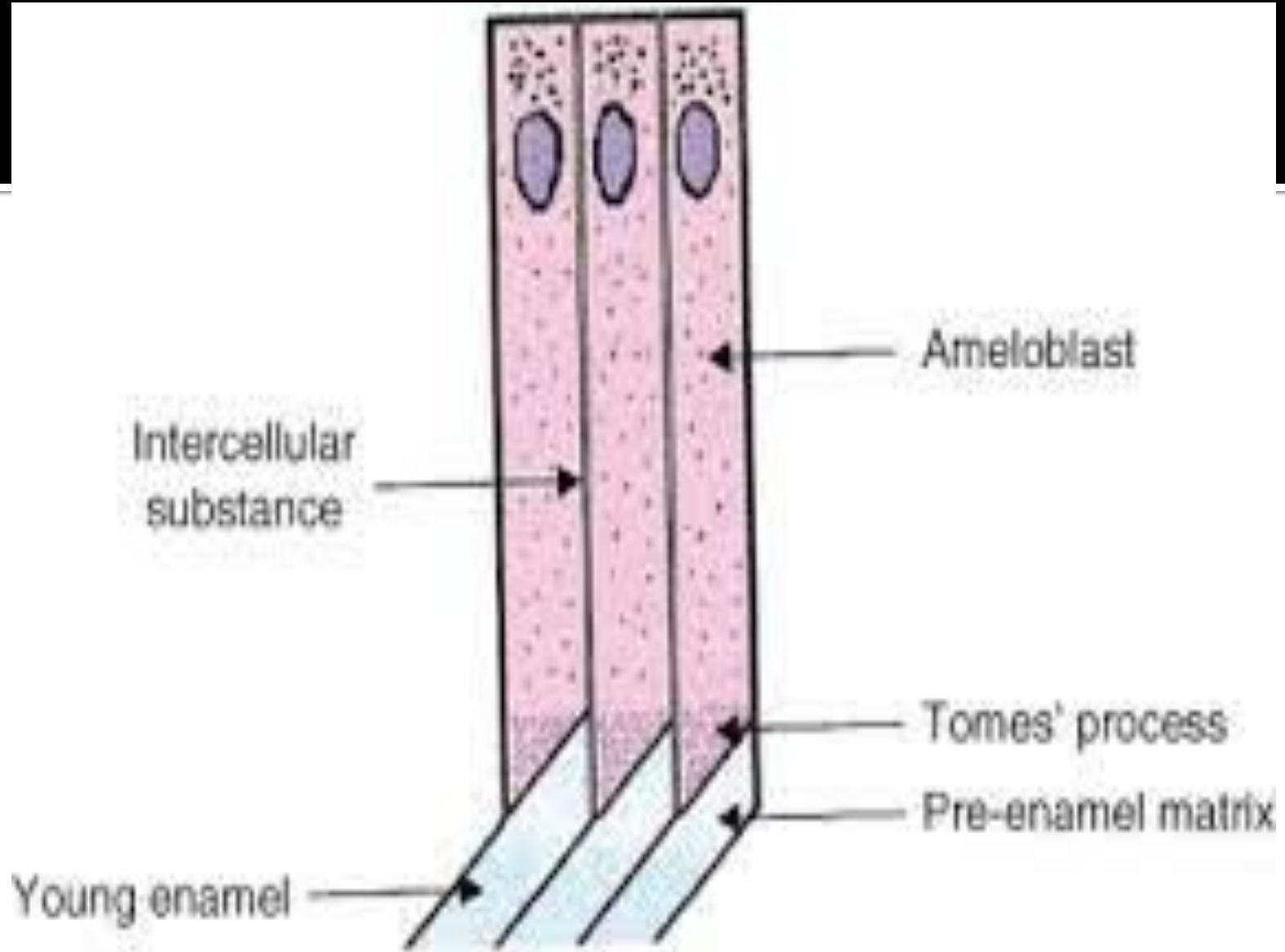


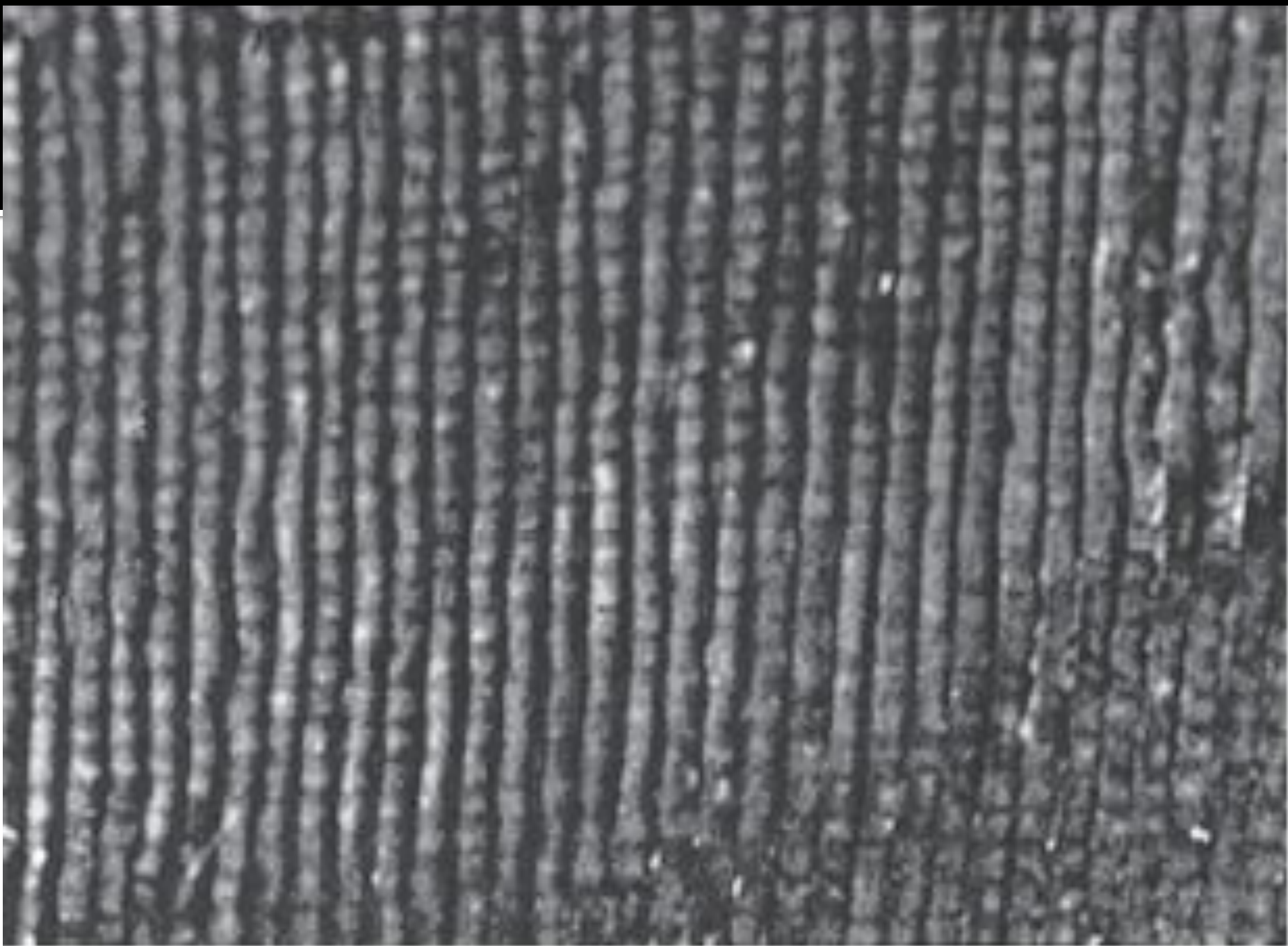
Aprismatic Enamel

- **The outer surface layer of enamel (20-100 μ m primary teeth and 20-70 μ m secondary teeth) is aprismatic.**
- **The crystallites are aligned at right angles to the surface and parallel to each other.**
- **The surface layer is more highly mineralised than the rest of the enamel. This is attributed to the absence of prism boundaries where organic material is located.**



- has also been reported to be present in the first enamel formed at the enamel–dentine junction.
- Nonprismatic enamel occurs as a result of the absence of Tomes processes from the ameloblasts in the first and final stages of enamel deposition





Box 7.1 Lines in enamel

Cross striations	Enamel stria	Neonatal line
Short period incremental lines	Long period incremental lines	Reflects metabolic changes at birth
Transverse to prisms	Oblique to prisms	Oblique to prisms
Throughout enamel	Throughout enamel	A single line
4 μm apart	25–40 μm apart	About 20–40 μm thick
Represent daily increments	Represent weekly increments	Represents a few days after birth
Prism varies in width	Reach surface at perikymata	Prisms vary in width and direction

Incremental line

- There are two main types of incremental line:
- short period (cross-striations) and long period (enamel striae, also known
- as the striae of Retzius
- In sections of enamel cut along the longitudinal axis of the crown the
- enamel striae (or striae of Retzius) are seen as prominent lines that
- run obliquely across the enamel prisms to the surface (Fig. 7.31).
In horizontal sections they form concentric rings (Fig. 7.32).
- They represent the successive positions of the enamel-forming front.
- The periodic nature of this feature, which may be assessed by the number of cross-striations between successive enamel

CROSS STRIATIONS

- Cross-striations are lines that cross enamel prisms at right angles to their long axis
- circadian clock
- at common interval of about 3–6 μm ,

STRIAE OF RETZIUS

- Enamel striae are also observed in demineralised sections
- enamel cut along the longitudinal axis of the crown the enamel striae (or striae of Retzius) are seen as prominent lines that run obliquely across the enamel prisms to the surf

STRIAE OF RETZIUS



Fig. 7.31 (A) Longitudinal section of enamel showing enamel striae running obliquely across the tissue. Wear at the tip of the cusp has exposed some of the striae on the surface at this site. Along the slopes of the tooth the striae naturally reach the surface ($\times 15$). (B) Higher-power view ($\times 40$) of enamel striae in enamel (A) along the side of the tooth running from the enamel–dentine junction (B) to the surface.

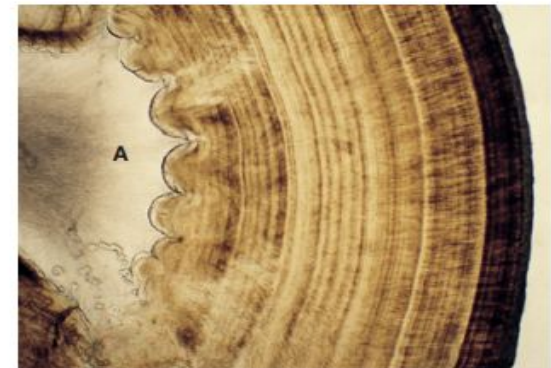
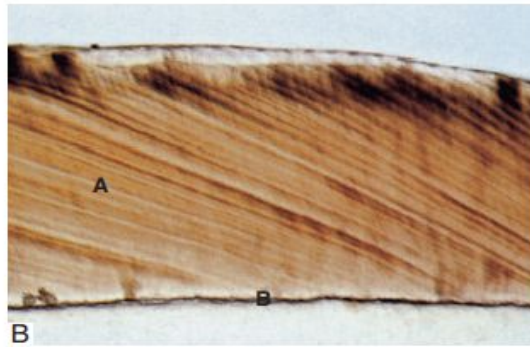


Fig. 7.32 Transverse section of enamel showing enamel striae running circumferentially. A = dentine ($\times 120$). Courtesy Dr M E Atkinson.

STRIA /PERIKYMATA

- In deciduous teeth, enamel striae and perikymata are only ever clearly
- seen in the cervical enamel of deciduous second molars.
- The exaggeration of striae in different teeth forming at the same time
- suggests a common systemic influence. One hypothesis is that there may
- be a rhythm with a 27-h cycle in addition to a diurnal daily rhythm. The
- two rhythms would coincide approximately every 7 or 8 days, producing
- a fault in the developing enamel

- Enamel striae are less pronounced or absent from enamel formed
- before birth. A particularly marked stria is formed at birth – this is the
- neonatal line and reflects the metabolic changes at birth.

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PERIKYMATA

- Over the whole of the lateral enamel, enamel striae reach the surface in a series of fine grooves running circumferentially around the crown.
- These features are known as the perikymata grooves and are separated by ridges, the perikymata ridge
- In deciduous teeth, enamel striae and perikymata are only ever clearly seen in the cervical enamel of deciduous second molar

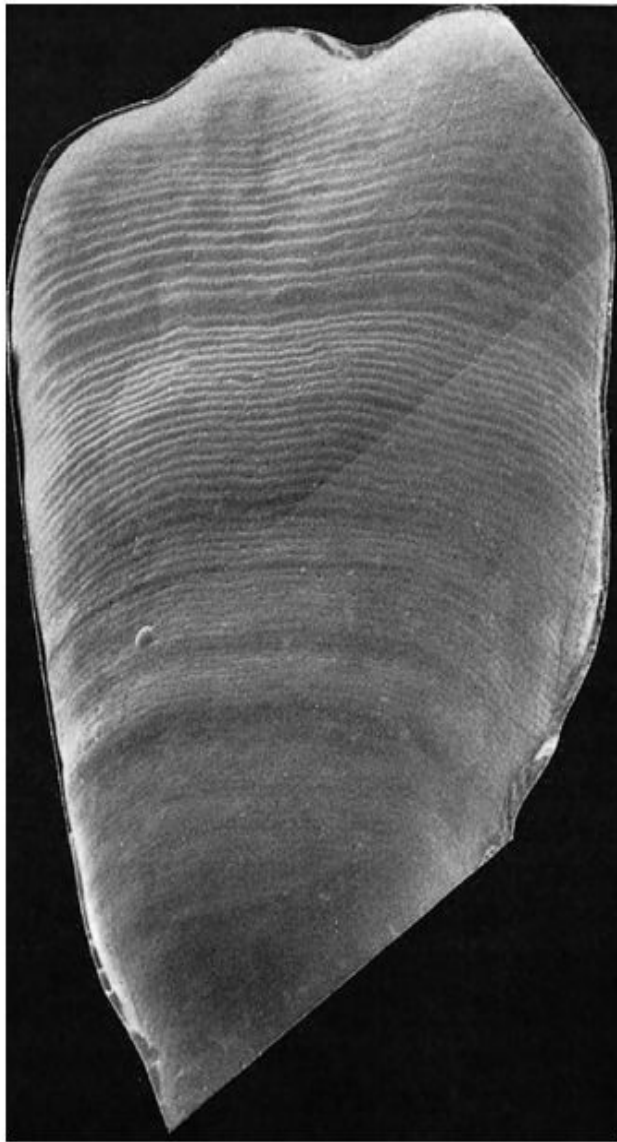


Fig. 7.37 SEM of enamel surface of a human lower incisor showing numerous transversely running perikymata grooves and ridges ($\times 10$). Courtesy Prof. M C Dean and the *Journal of Human Evolution*.

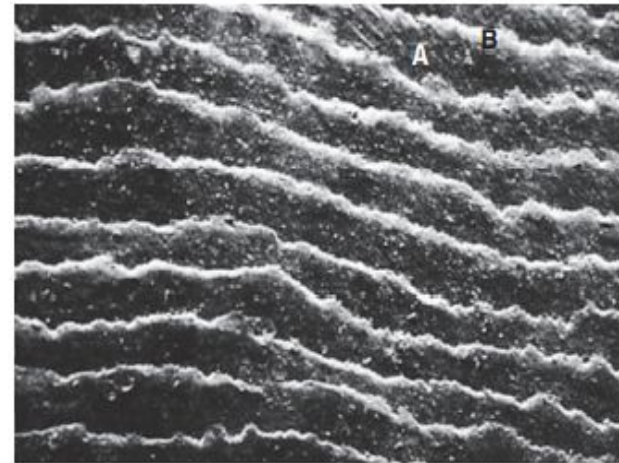


Fig. 7.38 Higher power view of surface enamel with perikymata ridges (A) separated by perikymata grooves (B) (SEM $\times 400$). Courtesy Dr D F G Poole.

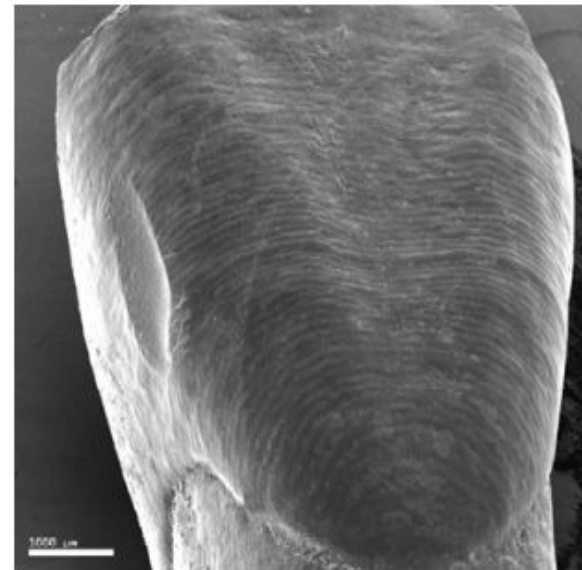


Fig. 7.39 SEM showing perikymata on the surface of a mandibular incisor from *Paranthropus robustus*. Marker bar = 1 mm. Courtesy Prof. M C Dean.

Attrition



Fig. 7.48 Attrition in patient suffering from bruxism (tooth grinding). Note the flattened occlusal plane. Courtesy

AGE CHANGES/



Fig. 7.48 Attrition in patient suffering from bruxism (tooth grinding). Note the flattened occlusal plane. Courtesy Prof. T J Watson.



Fig. 7.50 Erosion of the teeth associated with the consumption of grapefruit producing irregular concavities on the occlusal surfaces. Courtesy Prof. T J Watson.



Fig. 7.49 (A) Toothbrush abrasion in the cervical region of the labial surfaces of the anterior teeth. (B) Side view of abrasion. Although this appearance is usually attributed to frictional contact with an outside material such as dental floss or a toothbrush, other factors such as occlusal stress or corrosion from bacterial plaque may also be involved. The term 'abfraction' is used to describe this kind of lesion with a multifactorial basis. (A) Courtesy Prof. T J Watson. (B) Courtesy Dr J Grippo.



Age changes in enamel

- Enamel wears away slowly with age,
- diet and masticatory habits.
- darken in colour, (secondary dentine)

ENAMEL CARRIES

- Depending on the outcome, the initial lesion may be successfully halted and reversed by
- remineralisation or increase in size, setting up a vicious cycle of undermining the tooth until it reaches the dental pulp!

USE Of Substance to ARREST Caries

- Fluoride(drinking water, toothpaste or mouthwashes,)
- Calcium and phosphate
- Casein phosphopeptides(dairy products) stabilise calcium and phosphate ions.
- Bioactive materials-bioglass forms hydroxycarbonate

MICROPOROSITY OF ENAMEL

- In vivo, the pores in enamel are water-filled spaces between the crystallites.
- pore volume has been quoted as high as 10–12%
- Within the prisms most pores exist as very narrow gaps between closely packed crystallites
- pores are accessible only to small molecules such as water.